



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2025-26/operatio/382845

Date:

12/08/2025(18:31:57)

Subject:Approval for Revision of LMI on Operating Guidelines for Biomass Pellet co- firing in PF Boilers

Local management Instructions (LMI) for "Operating Guidelines for Biomass Pellet co-firing in PF Boilers" at NTPC simhadri has been reviewed and revised vide COS-ISO-00-OGN/OPS/SYST/049.

- The LMI has been revised as per new OGN , "COS-ISO-00- OGN/OPS/SYST/049 Rev. No.: 02, JUNE 2025".
- Submitted for the Kind approval of BUH please.

Attachements:

1. COS-ISO-00-OGN/OPS/SYST/049.
2. Revised Simhadri LMI.(word file)
3. Revised Simhadri LMI.(PDF file)

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The LMI document may please be rechecked and modified.

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LMI is in line with the OGN. Forwarded for further review.

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Operating Guidelines for Biomass Pellet co-firing in PF Boilers

COS-ISO-00-OGN/OPS/SYST/049 Rev. 02 July. 2025



NTPC Limited

ECM Ref No : SMTTP/OPERATION/2025-26/LMI/ 382845
Approved for Implementation by: HOP (Simhadri)

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LMI APPROVAL DOCUMENTS

Table of Contents

1.0	INTRODUCTION	4
2.0	SUPERSEDED DOCUMENTS	5
3.0	RECORDS OF REVISION	5
4.0	TYPES AND PROPERTIES OF BIOMASS PELLETS	7
5.0	HANDLING, STORAGE AND BLENDING OF PELLETS:	8
6.0	MONITORING OF CHEMISTRY PARAMETERS:	10
7.0	IMPACT OF BIOMASS CO FIRING ON COMBUSTION:	12
8.0	UNIT OPERATIONAL ISSUES WHILE HANDLING PELLETS:	14
9.0	ACTIONS TO BE TAKEN IN A MILLING SYSTEM HAVING FIRE DURING BIOMASS FIRING.	16
10.0	COMBUSTION ISSUES IN PELLET FIRING	17
11.0	IMPACT OF ASH AND USAGE IN CEMENT INDUSTRY	17
12.0	SAFETY ASPECTS OF PELLET FIRING:	17
13.0	INFRASTRUCTURAL REQUIREMENTS OF BIOMASS PELLET HANDLING	17
14.0	SITE SPECIFIC SOP'S TO BE PREPARED AGAINST THIS OGN	18
15.0	FURTHER ACTIONS REQUIRED	18
16.0	COMBUSTION AND CLINKERING ISSUES	18
17.0	REVIEW	18
	ANNEXURE -1-BIOMASS PELLET FIRING ENABLING AND INFRASTRUCTURE CHECKLIST	19
	ANNEXURE -2 - PARAMETER CHECKLIST	20
18.0	Annexure-3 - Logic for controlling Mill Inlet and Outlet temperature	21

1.0 INTRODUCTION

Biomass is an important energy source. Biomass is any organic matter—wood, crops, seaweed, stubble, animal wastes—that can be used as an energy source. Biomass is probably our oldest source of energy after the sun. For thousands of years, people have burned wood to heat their homes and cook their food. Biomass gets its energy from the sun. All organic matter contains stored energy from the sun. Biomass is a renewable energy source because its supplies are not limited. Stubble is the straw type material that remains after grains, like paddy, wheat, etc., have been harvested. Stubble burning is intentionally setting fire to the straw stubble that remains after grains, like paddy, wheat, etc., have been harvested to clear agricultural residue that is left on the land, to ready it for the next round of seeding. Stubble burning in fields emits large amount of ash/soot/ unburnt carbon to the atmosphere which is the real cause of air pollution and elevates the PM 2.5 and PM 10 level in atmosphere. It is learnt that farmers burn stubble as they find it as cheapest, quickest and easiest means to prepare the land for next crop in view of short time available. Stubble burning in fields also reduces soil fertility.

But, this stubble is a good biomass resource which has the potential to create efficient biomass-to-energy chains. Torrefaction of biomass stubble, combined with densification (Pelletisation or briquetting), is a promising step towards overcoming the logistical challenges in developing large-scale sustainable energy solutions, by making it easier to transport and store. Pellets or briquettes have higher density, contain less moisture, and are more stable in storage than the biomass they are derived from.

When agro residue based fuel, in the form of pellets, utilized in coal fired power plants, it burns completely in power plant and ash emitted from its combustion gets absorbed in Electro Static Precipitator (ESP) which prevents the air pollution while generating power from it.

Majority of power plants of NTPC are running on coal. To reduce greenhouse gas emission from its coal based power plants, NTPC intends to utilize agro residue based pellets/torrefied pellets along with coal for power generation through biomass co-firing which is a technology recognized by UNFCCC to mitigate carbon emission. It is worth mentioning that equivalent amount of CO₂ (carbon-di-oxide) emitted from combustion of agro residue based pellets/ torrefied pellets in power plant gets absorbed in next crop cycle by photosynthesis, so CO₂ emission from agro residue based pellets combustion does not increase CO₂ concentration in atmosphere and thus it is also termed as carbon neutral fuel which is a renewable source of energy. Further, CO₂ emission from diesel and electricity consumption for agro residue collection, processing and transportation is very negligible as compared to saving in CO₂ emissions from its utilization in large coal fired power plants having higher efficiency which makes biomass co-firing a greener alternative.

In addition to reducing carbon emission from coal based power plant, the utilization of agro residue based pellets/ torrefied pellets in the power plant will also reduce air pollution due to burning of stubble (i.e. paddy straw and other agro residues) in the fields by farmers which was the trigger for this initiative

by NTPC as a responsible organization. Emissions of sulfur and mercury are reduced by the co firing percentage.

2.0 Superseded Documents –

- SMTTP/OPERATION/2025-26/LMI/874578
- OGN/OPS/SYST/049 Rev. No. 1, Aug. 2023

3.0 Records Of Revision

Clause No.	Description	Rev. No.01	Rev. No. 02
4.3	Types and Properties of biomass pellets: Biomass Blend with Coal	-	New Clause Added Blending Biomass with coal makes the mix comparatively safer to fire in boiler as compared to biomass.
6.0	Sample collection	Sample collection is done from truck-top. Top 25 cm is removed and then randomly samples are collected from 4-5 spots.	The sample is to be collected from each truck/consignment from 4-5 randomly selected spots (from any location to obtain representative sample).
8.0 (v) (C)	Unit Operational issues while handling pellets:	Mill inlet temperature set point should be controlled in such a way that operator cannot give more than 180 OC set point for mills with bunkers fed with non torrefied biomass pellets, however in case of torrefied biomass pellets cofiring the setpoint to be restricted below 210 deg.C.	Mill inlet temperature set point should be controlled in such a way that operator cannot give more than 240 OC set point for mills with bunkers fed with biomass pellets(torrefied & non torrefied). It is to be noted that blending ratio must not exceed 10%.

8.0 (a) (v) (d)	Unit Operational issues while handling pellets:	In case of Torrefied biomass pellets cofiring: Mill HAG (hot air Gate) shall close on protection if Mill inlet temperature reaches 220 deg C. Alarm for Mill inlet temperature to be provided at 215 deg C	In case of biomass pellets cofiring: Mill HAG (hot air Gate) shall close on protection if Mill inlet temperature reaches 250 deg C. Alarm for Mill inlet temperature to be provided at 245 deg C.
8.0 (a) (v) (f)	Unit Operational issues while handling pellets:	Alarm for mill outlet temperature >75 deg C to be provided.	Alarm for mill outlet temperature >70 deg C to be provided.
8.0 (a) (v) (g)	Unit Operational issues while handling pellets:		New Point added Mill outlet temperature setpoint must not be increased beyond 65 Deg. C and Protection close to HAG should be kept at 75 deg. C.
8.0 (b) (i)	Unit Operational issues while handling pellets:	Maintain Mill outlet temperature ≥ 55 deg C	Maintain Mill outlet temperature ≥ 57 deg C and less than 65 deg. C
8.0 (b) (ii)	Unit Operational issues while handling pellets:	Mill inlet temperature should be ≤ 180 deg C in case of non-torrefied pellets cofiring	Mill inlet temperature should be ≤ 240 deg C in case of non-torrefied/Torrefied pellets cofiring
8.0 (b) (iii)	Unit Operational issues while handling pellets:	Mill inlet temperature should be ≤ 210 deg C for torrefied pellets cofiring.	A logic for controlling mill inlet and outlet temperature is attached in annexure-3
8.0 (b) (viii)	Unit Operational issues while handling pellets:	Mill inlet temperature should not be more than 180 deg C and 210 deg C for non -torrefied and torrefied pellets cofiring respectively.	Point deleted. It is covered at 8.0 (b) (iii)
4.3	added "However, individual station may send source wise coal sample along with biomass pellets for any change in properties with respect to different blending ratio".		

6	Changed for testing of parameters.		
13	added "Proper PPE's to be used during sample collection, preparation and testing."		
15	added "Sample to be sent to NETRA for Ultimate analysis, Chlorine and alkali metals."		

4.0 Types and Properties of biomass pellets

There are three types of pellets:

- Non torrefied biomass pellets
- Torrefied biomass pellets
- Biomass blend with Coal.

4.1 Non-Torrefied biomass pellets are the pellets made from agro based residue without torrefication. The main disadvantage with these pellets is that they are hygroscopic in nature and absorbs moisture readily.

The typical properties of non torrefied pellets are as follows:

- Carbon Content: 10-20 %
- Volatile Matter: 60-66%
- Moisture: 8%
- Density: 700 kg/m³
- Ash content : approx. 20%
- GCV: 3500 – 3800 Kcal/kg

These properties are indicative in nature, however the properties of pellets should be taken from torrefied pellets procurement terms and conditions.

The main constraints with these pellets are

- Ignition temperature: 240⁰ Celsius
- Moisture affinity : Very high

In order to overcome these constraints, torrefaction process will greatly help.

4.2 Torrefied biomass pellets:

Torrefaction is a thermal process to convert biomass into a coal-like material, which has better fuel characteristics than the original biomass. Torrefied biomass is more brittle, making grinding easier and less energy intensive. Compared to fresh biomass, storage of the torrefied material can be substantially simplified since biological degradation and water uptake is minimized. Torrefaction involves the heating of biomass in the absence of oxygen to a temperature of typically 200 to 400°C. The structure of the biomass changes in such a way, that the material becomes brittle, and more hydrophobic. Although the weight loss is about 30%, the energy loss is only 10%. Main product is the solid, torrefied biomass. During the torrefaction process a combustible gas is released, which is utilized to provide heat to the process.

4.3 Biomass Blend with Coal:

Blending Biomass with coal makes the mix comparatively safer to fire in boiler, As per NETRA the properties of Blend is shown below:

Sl. No.	% Biomass	% Coal	De-volatization temp (Deg. C)	Ignition temperature (Deg. C.)
1	0	100	360	407
2	10	90	293	402
3	20	80	282	396
4	30	70	268	391
5	100	00	220	230

source wise coal sample along with biomass pellets for any change in properties with respect to different blending ratio has to be sent to NETRA. **–Responsibility Chemistry**

5.0 Handling, storage and blending of pellets:

–Responsibility CHP

- a. The following steps are involved in handling
 - i. Receipt
 - ii. Unloading
 - iii. storage
 - iv. Feeding to Bunkers
- b. Receipt and unloading:
 - i. The truck containing Biomass pellets (app 20 T/day) enters plant premises
 - ii. Vendor test report accompanied the lot shall be checked.
 - iii. Sample collection and result at station end by chemistry group.
 - iv. Necessary arrangements for safe sample collection like platforms etc., are to be ensured at site.

- v. Acceptance of the lot based on the result given by the chemistry group
 - vi. After acceptance, Truck shall be weighed.
 - vii. After weighment truck to be unloaded at designated unloading point.
 - viii. Truck unloading mechanism: truck unloader should be available at site if more quantity of pellets are used for blending.
 - ix. Empty truck shall be weighed.
 - x. After weighment empty truck to be released from plant premises.
- c. Storage:
- i. Biomass pellets area for storage needs to be identified by the site. This area should have the facilities for firefighting.
 - ii. The volatile matter in biomass pellets is very high and can easily catch fire. Hence continuous monitoring of the area needs to be done.
 - iii. The pellet storage area should be different from coal stockyard.
 - iv. Facilities should be developed to transport the biomass pellets from storage area to blending area.
 - v. The maximum percentage of blending allowed is 10% for non-torrefied pellets and 20% for torrefied pellets depending on plant operating conditions and combustion issues (blending by heat value).
 - vi. The bunkers in which blended coal with biomass pellets needs to be identified along with operation group and all necessary interlocks to be ensured before dumping coal in bunkers.
 - vii. Avoid giving Hot work permits on/near pellets feeding path.
 - viii. Don't spray water / dry Fog / plain water fogging system on the pellets and its bunker feeding path. The biomass pellets are hygroscopic in nature. After absorbing water or moisture, the pellets loses its shape and converts to powdery form, so water cannot be used. However, the system for dust suppression like dry fog/plain water and spray water should be available and for use in case of emergency.
 - ix. Don't compact the pellets.
- d. Bunkering and blending

- i. The maximum percentage of blending allowed is 10% for non-torrefied pellets and 20% for torrefied pellets depending on plant operating conditions and combustion issues (blending by heat value).

Note: Blending to be restricted to 5% in case of Ball and tube mills and only Torrefied pellets to be blended with VM below 22%.

- ii. Site specific Bunkering methodology is to be formulated by considering the following points.
 - 1. Point of blending
 - 2. Methodology of feeding pellets
 - 3. TP or chute where blending shall be done.
 - 4. Control methodology for blending like using belt weigh scales etc.,
 - 5. No of bunkers to be used for pellets blending
 - 6. Level to be maintained in bunkers in which pellets are bunkered
- iii. The pellets do not require crushing, hence blending is to be done after crusher output.
- iv. Feeding to be done only along with the coal, as the biomass pellets are highly inflammable as it has very low ignition point and high VM content. Hence conveyor interlock to be modified so that pellets go only when coal is in conveyor, and not otherwise.
- v. Conveyor streams and chutes used for pellet firing must be thoroughly emptied after feed operations. Left over pellet blends in conveyors and chutes must be emptied to minimize fire hazards.

6.0 Monitoring of chemistry parameters:

–Responsibility Chemistry

Biomass Pellets:

a. Sample collection:

- i. To be done for each and every truck arriving at the station.
- ii. The sample is to be collected from each truck/consignment from 4-5 randomly selected spots (from any location to obtain representative sample).
- iii.

b. Sample Preparation:

- i. The collected sample is brought immediately to Chemistry lab premises and all portions collected are thoroughly mixed followed by coning-quartering until the sample quantity reduces to approximately 4 kg.
 - ii. This 4 kg sample is divided into 4 equal parts- one part for NTPC, one part for seller, third part for referee purpose and the fourth part is used for fineness analysis and particle size distribution.
- c. Sample analysis:

Sl.No	Description	Applicable Standard	Remarks
1	Moisture (*ARB) analysis	ISO 18134 (P3)/ IS 17655 (P3)/ Equivalent method/Hand-held Moisture meter	To be tested before unloading and rejected if exceeds the rejection limit (as per contractual terms and conditions).
2	Sample collection and preparation	ISO 18135/ IS 18640	• In presence of representatives of NTPC and pellet supplier. • Single sample shall be prepared for all the consignments received in a day from a particular vendor for testing of GCV and other parameters. • In case of supply from multiple vendors in a day, vendor-wise samples shall be prepared for all the consignments received in a day. • Sampling: The sample quantity of approximately 30 Kg is to be collected from each truck/consignment from 4-5 randomly selected spots (from any location of the truck) after removing biomass pellet layer of approximately 25-30 cm depth from the top / or anywhere from the truck as decided by Station (to obtain a representative sample). Approximately 5-6 kg of sample is to be collected from each spot.
3	Fines % (*ARB) (Length< 3 mm) determination	ISO 18846/ IS 17656/ equivalent method	If > 5% (or as per contractual terms and conditions), proportionate recovery shall be done.
4	Gross Calorific Value (*ARB), kcal/kg	ISO 18125/ IS 17654/ Equivalent method	Penalty as per contractual terms and conditions
5	Volatile Matter (*ARB) (%)	ISO 18123/ IS 17844	Penalty (if any) as per contractual terms and conditions

6	Sulphur (*ARB) (%)	ISO 16994/ IS 17833/ Equivalent Method	Penalty (if any) as per contractual terms and conditions
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*** ARB: As-Received Basis**

7.0 Impact of biomass co firing on combustion:

- a. The process of biomass combustion may be associated with certain risks that do not occur during combustion of coal. These include fuel pre-processing (fire-explosion risk), combustion eg including excessive slagging and ashing, and chlorine corrosion. Therefore, knowledge of the physicochemical properties of plant biomass helps to determine its potential application in heat or electricity production. The knowledge of these parameters allows proper selection of the amount of combusted biomass to ensure its minimal impact on the boiler system or the use of preventive measures minimizing the negative impact of biomass combustion.
- b. Content of Oxygen in biomass is very high compared to coal.
- c. The substantial proportion of volatile matter in the biomass fuel can be a positive factor in improvement of ignition and flame stability. However, volatile matter enhances the fire-explosion risk in the pre-processing system.
- d. Biomass pellets may have very low content of sulphur and nitrogen compared to coal which makes them environmental friendly by reduced SOX level.
- e. Burning biomass fuels or biomass-coal mixtures containing low sulphur content is valuable for major reduction of Sox /SO₂ emissions but might negatively influence the ash deposition behaviour, in particular Chlorides deposition. It has been generally accepted that the occurrence of Sulphur can alleviate corrosion problems associated with chloride deposits via the following sulphation mechanism.
- f. In biomass, elements such as chlorine and potassium are mostly present as water soluble inorganic salts, and primarily as chloride, nitrates, and oxides etc. which can be easily volatilized during the combustion, resulting in high mobility for alkali materials and, consequently, high pollution tendency.
- g. An important element in the use of biofuels in the power industry, in particular in the combustion process, is their adverse effect on the slagging and fouling processes. The chemical composition of biomass ash is significantly different from that of coal. The mineral material from biomass does not contain aluminosilicates but does contain quartz and simple inorganic salts of potassium, calcium, magnesium, and sodium in the form of phosphates, sulphates, and chlorides.

- h. One area of particular concern is ash composition of biomass fuels. Most of these ashes are high in alkali and alkaline earth elements which are known to promote deposition in the boiler. Even though the ash quantity in the biomass is small, high alkali and alkaline earth materials tend to act as fluxing agents and reduce the melting temperature of the coal ash. This should be evaluated. One characteristic feature of the ashes yielded by biomass combustion is the considerable content of phosphorus, potassium and their compounds, calcium, magnesium etc. Biomass ash is thus characterized by a low melting temperature and a high tendency to slagging and fouling. Ca and Mg compounds usually increase the ash melting temperature, while K and Na reduce it. In combination with potassium, silicon can induce formation of low-melting silicates in volatile ash particles. These processes are highly important, given the risk of fouling and ash slagging on the walls of furnaces or heated surfaces.
- i. The main effect of biomass co-firing with coal is the emission of vapours of potassium compounds and subsequent condensation on the surface of ash particles and boiler pipes.
- j. The two most abundant alkali metals contained in the fuel are usually potassium and sodium. They exist in the original fuel in different forms. These alkali species are mostly volatilized with organic species, or may be released as metallic elements, although they then rapidly react. If fuel contains chlorine, alkalis probably appear as the salts KCl or NaCl. Some of the sodium and potassium are readily volatile even at relatively low temperatures. However, on the particle surface, where combustion takes place, temperatures may be considerably higher than the overall bulk gas temperature; this would suggest that most alkalis are vaporized during combustion and are available for reaction with other compounds. For example, sodium and potassium may react with SO₂ or SO₃ in the gas to form the alkali sulphates, K₂SO₄ and Na₂SO₄, which can condense and deposit. In this manner, the surface initially acquires a characteristic thin, dense, and reflective deposit layer observed in nearly all cases. The vaporization and subsequent chemical reactions are responsible for much of the fouling, corrosion, and silicate formation found in boilers. Addition of these alkalis tends to lower the fusion temperatures of the solids, making them more liquids and able to stick to water wall surfaces.
- k. One method of estimating the amount of alkali readily volatilized is to determine the “water soluble alkalis.” This test is intended to measure the quantity of alkali present in the form of compounds that may be readily volatilized during combustion. Most sodium and potassium compounds are water soluble; the assumption is that those that are not water soluble are present in the form of large complex molecules such as clays and would not readily decompose on

heating. The greater the fraction of water-soluble alkalis, the greater will be the potential for deposition.

- l. Role of Calcium: Calcium and other alkaline earth materials are found in coal. Calcium in fuel is more refractory and less readily volatilized, although some may be released by similar mechanisms and in similar forms as the alkalis. In general, alkaline earth species form more stable compounds that are less volatile than materials formed from alkali materials. In addition, calcium may react with gaseous sulphur species and remove them from contributing to formation of low-melting compounds.
- m. Increased levels of potassium compounds in biomass are a highly unfavourable phenomenon due to the process of slagging and clinker formation. These components are characterised by formation of low-melting eutectics, which contribute to a faster rate of ash melting and deposition on superheater tubes and heating surfaces.
- n. Chlorine is an undesirable element in the process of fuel combustion due to the increased risk of high-temperature chlorine corrosion. The enhanced risk of corrosion is associated with the chemistry of ash deposits accumulated on boiler surfaces. The surfaces of superheaters are mostly at risk due to the high temperature of the operating factor, which results in increased tendency of surface deposits to melt and react with the surface of super heater tubes. Changes in the ash composition and quality occurring during the co-firing process may also influence the use and storability of ashes. The methods applied in energy industry for purifying solid-derived exhausts in electrostatic precipitators or wet desulphurisation leads to formation of by-products, which are used in other sectors, i.e., construction and cement industries. An increase in the amounts of undesirable substances may result in difficulties in management thereof and a necessity to store them, which significantly reduces the efficiency of power plants.
- o. If fly ash and bottom ash are currently sold in the concrete market, an evaluation must be made since current standards are for coal ash only.

station should get the biomass pellets ultimate analysis done before start of the biomass pellets co-firing and understand the impact on the combustion based on the ultimate analysis. Ash elemental analysis is also to be done during the initial days of firing in the boiler and furnace temperature measurement is to be done frequently

during the initiate days of biomass co-firing. Boiler is to be monitored for ash build up and slagging during the biomass co-firing. Combustion has to be monitored closely while biomass co-firing takes place.

8.0 Unit Operational issues while handling pellets:

- a. The following systems are to be made ready before going for pellets operation.
 - i. Mill inerting system is charged up to mill inlet. The valve provided at mill inlet should be operable from UCR (either motorized or pneumatic) along with local pressure gauge after valve. **Responsibility -OPR/EMD/C&I**
 - ii. Orifice is provided in common header to maintain mill inerting steam pressure 02-04 ksc. - **Responsibility -BMD**
 - iii. All individual mill outlet pipes are to be provided with RTDs and the temperature of coal pipes to be hooked up to UCR. **Responsibility -C&I**
 - iv. Logic modifications as follows are required before starting the firing pellets in the mills. **Responsibility -C&I**
 - v. For operator's ease, Biomass ON/OFF buttons must be provided on individual Mill screen. This button shall be used for change over to Biomass mode and vice-versa. **Responsibility -C&I**
 - a. Whenever Biomass firing is ON, then following changes should happen in the Mill logics: Mill temperature control by CAD (cold air damper) of the mill to be shifted to Mill inlet temperature control as per the Inlet temperature set point. Mill inlet temperature set point increase/decrease button may be provided in control interface.
 - b. If biomass is being fired first time in that unit, Mill inlet temperature should be restricted to 180 Deg. C and should be increased in steps of 10 deg. After every 24 hours operation without abnormality.
 - c. Mill inlet temperature set point should be controlled in such a way that operator cannot give more than 240 °C set point for mills with bunkers fed with biomass pellets (torrefied & non torrefied). It is to be noted that blending ratio must not exceed 10%.
 - d. In case of biomass pellets cofiring: Mill HAG (hot air Gate) shall close on protection if Mill inlet temperature reaches 250 °C. Alarm for Mill inlet temperature to be provided at 245 °C.
 - e. Mill protection at mill outlet temperature of 95° C & 110° C will also remain in service irrespective of Biomass ON/OFF mode.
 - f. Alarm for mill outlet temperature >70 °C to be provided.
 - g. Mill outlet temperature setpoint must not be increased beyond 65 Deg. C and Protection close to HAG should be kept at 75 deg. C.
- b. **Mill Operating Parameters and Operation Procedure: -**
 - i. Maintain Mill outlet temperature $\geq 57^{\circ}\text{C}$ and less than 65 deg. C
 - ii. Mill inlet temperature should be $\leq 240^{\circ}\text{C}$ in case of non-torrefied/Torrefied pellets cofiring)

- i. In order to maintain mill inlet and outlet temperature a hybrid logic is to be put in mill temperature control, , typical logic for controlling Mill i/l and o/l temperature is attached as per Annexure-3.
 - iv. Observe all coal pipe temperature of Mills with biomass every shift. In case of any deviation from rest of the pipes (any abrupt rise or fall), stop the mill and give PTW for inspection of coal pipes.
 - v. Initially, periodic internal inspection of mills should be done during biomass co-firing to observe any possible accumulation of biomass inside mills. Initially, the periodic inspection may be carried out in shorter frequency say every 1 hour and gradually it may be increased based on experience. This is the vital aspect from the safety point of view.
 - vi. Prestart and post shutdown purging of mill is mandatory. In case of Mill Trip, the Mill must be emptied at the earliest to avoid fire in the mill.
 - vi. Continuous monitoring of mill parameters most importantly mill inlet air temperature (Mix Air Temperature), Mill outlet temperature, mill current, and mill DPs.
 - v i. Mill reject system to be kept under observation.
 - ix. Deployed Mill Operator must monitor for any kind of Fire in the rejects and intimate control Room immediately in wake of fire hazard.
 - x. In case of fire, immediate action to be taken as per OD "The Safe Operation and Maintenance of Pulverized Fuel Plant" OD/OPS/SYST/009 Latest revision.
 - xi. View glass of Mill feeders should be cleaned periodically, and seal air arrangement shall be provided to prevent dust deposition on view glasses. Illumination inside feeder shall be revived/ provided to keep track on biomass and coal blend.
 - xi. Regularly watch flue gas exit temperature and adjust SAPH/PAPH flue gas dampers, if required, so that increase in overall flue gas exit temperature can be minimized to the extent possible.
- c. Step by step procedure for going to biomass mode shall be provided to all desk engineers and proper training to these people essential before going for pellet firing. **Responsibility -OPR**

- d. Changes in the logbook and log sheets to be done before going for pellet firing and operator should monitor the parameters to and record them in the log sheets. - **Responsibility -OPR**

9.0 Actions to be taken in a milling system having fire during Biomass firing:

Responsibility -OPR

- a. Call the CISF fire personnel for immediate assistance.
- b. Follow the instructions given in OD-OPS-SYST-009 - The Safe Operation and Maintenance of Pulverized Fuel Plant.
- c. Site specific SOP to be prepared for handling mill fires in stations.
- d. Do not expose any part of the mill (roller, scrapper chamber and tramp iron gate for any inspection until the mill temperature comes down to normal value)

10.0 Combustion issues in pellet firing

- a. Clinkering and slagging tendency to be observed by any rise in SH zone FG temperatures. Frequency of Soot blowing and LRSB to be determined accordingly. **Responsibility -OPR**
- b. Follow the OGN-OPS-SYST-001 - Prevention and safe cleaning of Ash Build-up in Boiler Bottom Ash Hopper. **Responsibility -OPR**
- c. Flue gas temperature profile increased unburnt, changes in Spray and metal temperature to be monitored and any abnormality to be discussed with Corporate Operation services. **Responsibility -OPR**
- d. Elemental analysis of biomass for each sample/lot to be done to keep check on the chlorine content and alkali content like Sodium, potassium etc. which have high slagging and fouling tendencies due to sodium and potassium compounds formed and mixing with bottom ash. Similarly, chlorine content to be kept under control which has tendency of corrosion. **Responsibility -CHEMISTRY**

11.0 Impact of ash and usage in cement industry

Currently, the American Society for Testing and Materials (ASTM), ASTM Standard C618, does not allow ash from a co-firing installation to be used in the concrete arena. If fly ash from a plant is currently being sold to this market, that impact needs to be studied during co-firing.

12.0 Safety aspects of pellet firing:

Responsibility -CHP

- a. Pellets have got very high amount of volatile matter.
- b. Storage area must be predefined, and proper barricading is to be ensured. No cutting and welding works should be carried out near that area. Area should be clearly demarcated as “No smoking Area”.
- c. Conveyor firefighting system should always be healthy as the fuel is highly inflammable; hence all protection needs to be healthy.
- d. Storage area should have proper firefighting provisions like hydrants and water monitors.
- e. MSDS for these pellets are to be displayed near the storage area.

13.0 Infrastructural requirements of biomass pellet handling

Responsibility -CHP

- a. Truck tippler – 60 tons two nos.
- b. Closed shed (from sides also) covering feeding Area as well as Storage space as the fuel is highly susceptible to atmospheric moisture if left in open for reasonable amount of time. The Shed should cover the feeding point and the storage yard and should allow the movement of dozing equipment's like pay-loader or bob-cat. Site team shall decide the dimensions / Size and location of the shed. The shed should be provided with flame proof lighting.
- c. Weighbridges calibrated for weighing of the trucks.
- d. Belt weighing scales for proper blending.
- e. Pay loader or more preferably by Skid Steer Loaders for feeding and for achieving finer control over the feed rate.
- f. Safe Working Platform for collecting sample from truck.
- g. Methane, CO, Multi gas detectors at storage location
- h. CCTVs for monitoring at different points like unloading Points, Blend points, Feed points and conveying path.
- i. Proper PPE's to be used during sample collection, preparation and testing

14.0 Site specific SOP's to be prepared against this OGN

- a. SOP for unloading, storage and bunkering and blending for use in CHP with all precautions to be taken. **Responsibility -CHP**
- b. SOP for main plant operation activities from Mills to ESP including parameters to be monitored, Emergency instructions and logic modification done etc. -
Responsibility -OPR / C&I
- c. SOP for chemistry for sample preparation to analysis. **Responsibility -CHEMISTRY**

15.0 Further actions required:

- a. Ultimate analysis of biomass pellets to be done. **Responsibility -CHEMISTRY**
- b. Ash elemental analysis after biomass firing and ash fusion temperature monitoring. - **Responsibility -CHEMISTRY**
- c. Impact of biomass on the quality of ash as required for cement industries.
- d. Effect of pellet-ash on performance of wet / dry ash handling system to be checked periodically by stations. (Like scaling inside pipes, hoppers, sumps, silos etc. and evacuation & flowability of dry ash from hoppers)
- e. All parameters impacting heat rate shall be recorded and Heat rate before start of biomass co-firing and during biomass co-firing should be recorded to analyse its commercial impact. - **Responsibility -EEMG**
- f. Sample to be sent to NETRA for Ultimate analysis, Chlorine and alkali metals

16.0 Combustion and clinkering issues

- a. An important element in the use of biofuels in the power industry, in particular in the combustion process, is their adverse effect on the slagging and fouling processes.
- b. Units with clinkering tendency to be put last on priority for biomass firing.
- c. Mill capacity is reduced in biomass firing therefore it should be kept in consideration when deciding extant of biomass blending.
- d. PA header pressure to be kept 50 mm higher to avoid mill settlement.

17.0 Review

The Head of Station Operation Services will be responsible for reviewing this Guidance Note on a two-yearly basis or as necessary.

18.0 Annexure -1-Biomass pellet firing enabling and infrastructure checklist.

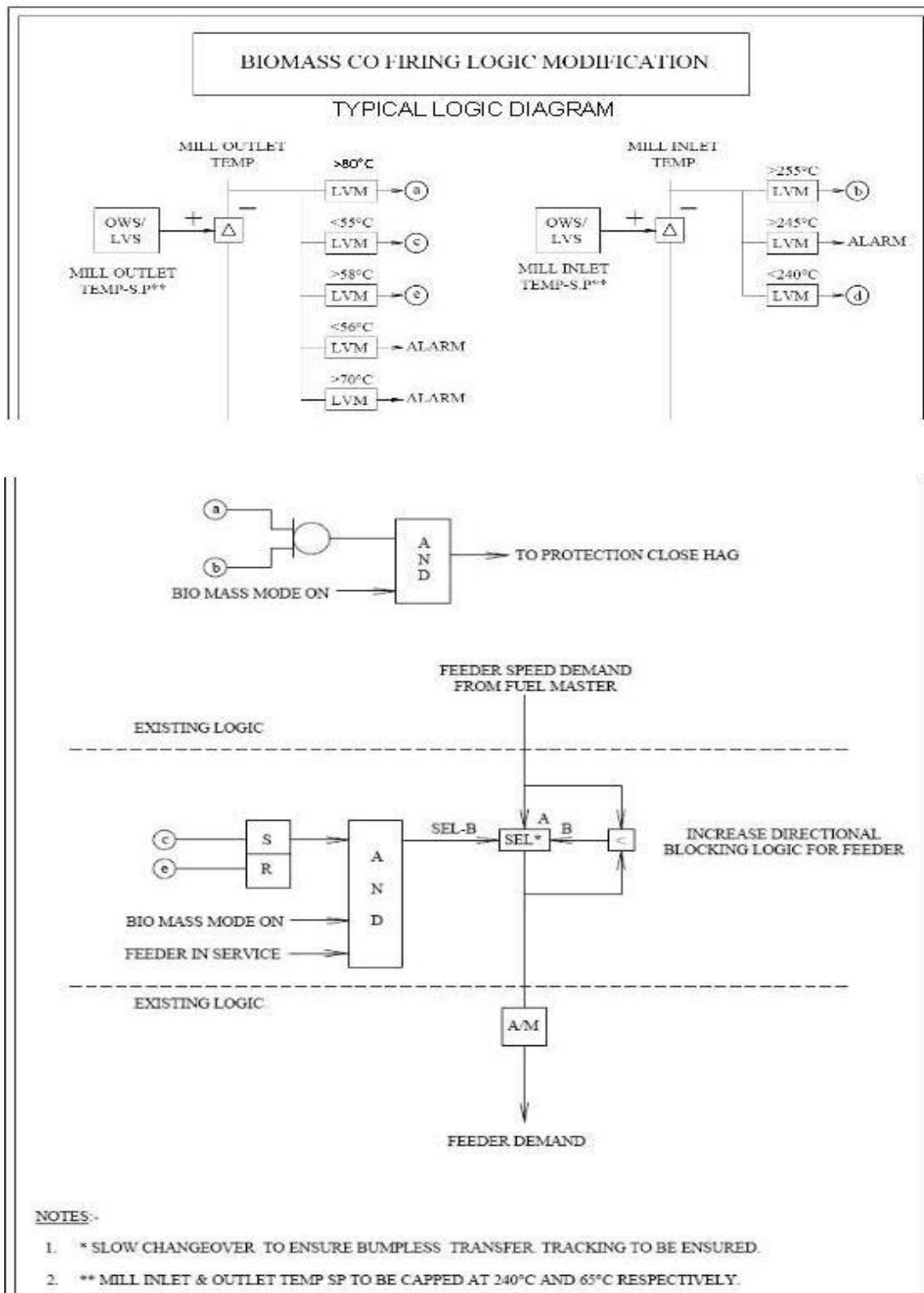
S.NO.	SUBJECT	Compliance Status
1.	Preparation of LMI on “Biomass pellet firing”	Yes/No
2.	Preparation of SOP on “Biomass pellet firing” by Opn	Yes/No
3.	Preparation of SOP on “Biomass pellet firing” by CHP	Yes/No
4	Preparation of SOP on “Biomass pellet firing” by Chemistry	Yes/No
5.	Pellet storage area different from coal stockyard?	Yes/No
6.	Maximum Percent blending done with coal?	
7.	Bunkers identified for pellet feeding? Please give details of mills identified here.	Yes/No
8.	Furnace temperature measurement: 1. Without biomass pellet firing 2. With biomass pellet firing	
9.	Mill inerting system charged upto mill	Yes/No
10.	action plan if any for converting Mill inerting valve to remote operation valve	Yes/No
11.	All individual mill outlet pipes are provided with RTDs and the temperature of coal pipes available in UCR.	Yes/No
12	Logic modification done as per OGN or not If no action plan for doing the same	Yes/No
13	a. Biomass firing ON/OFF toggle	Yes/No
	b. Mill i/l temp. control	Yes/No
14	Infrastructure available	
	1. Truck tippler – 60 tons two nos If not available, action plan available 2. Closed shed (from sides also) covering feeding Area as well as Storage space as the fuel is highly susceptible to atmospheric moisture if left in open for reasonable amount of time. The Shed should cover the feeding point and the storage yard and should allow the movement of dozing equipment's like pay-loader or bob-cat. Site team shall decide the dimensions / Size and location of the shed. The shed should be provided with flame proof lighting Action plan if not there? 3. Weighbridges calibrated for weighing of the trucks 4. Belt weighing scales for proper blending 5. Pay loader or more preferably by Skid Steer Loaders for feeding and for achieving finer control over	Yes/No

	the feed rate. 6. Safe Working Platform for collecting sample from truck. 7. Methane, CO, Multi gas detectors at storage location 8. CCTVs for monitoring at different points like unloading Points, Blend points, Feed points and conveying path.	
15	Sample of first lot of biomass elemental analysis done from NETRA or not If sample is not yet analysed, Please send the sample and get complete analysis done	Yes/No
16	Any incidents of combustion problems like clinkering and slagging? Capture date and time with data for further analysis	Yes/No

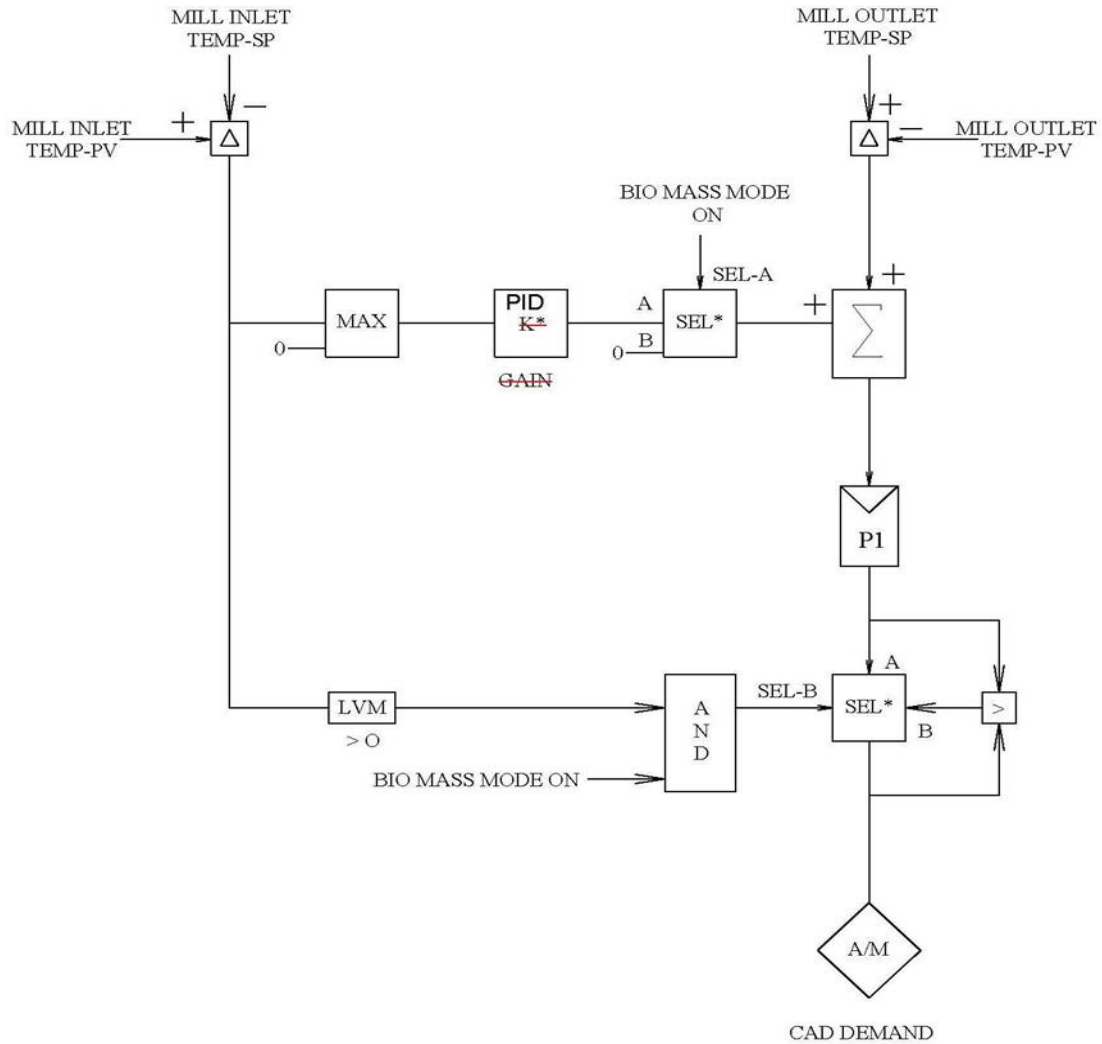
Annexure -2 - Parameter Checklist

S.No.	Parameter	Value Time:.....
1	Bunkers in which pellet being fed	
2	Mills in pellet firing mode	
3	Mill inlet temperature setpoint/Actual	
4	Mill inerting system available	
5	Mill current	
6	Mill DP	
7	Mill outlet temperature	
8	Mill feeding	
9	Coal pipe temperatures	
10	Mill Reject clear	Yes/No
11	FEGT	
12	HAD/CAD in auto	Yes/No
13	Bottom ash unburnt	Last value and time
14	PA Header pr setpoint	
15	Any clinkering/Slagging tendency	

Annexure-3 - Logic for controlling Mill Inlet and Outlet temperature



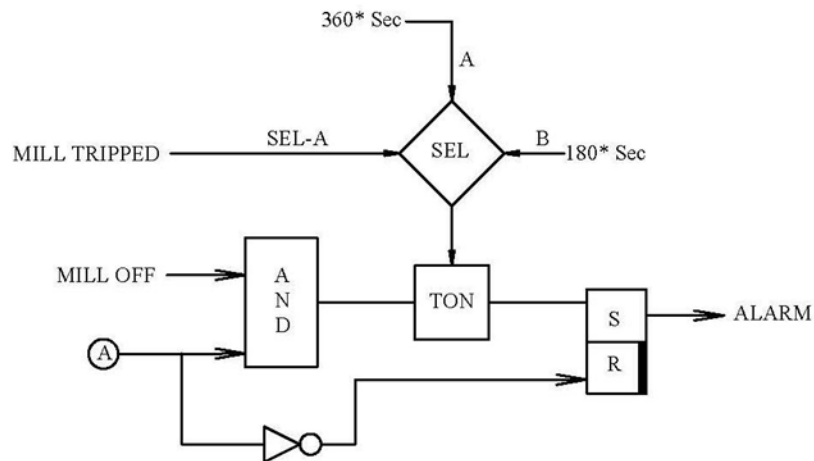
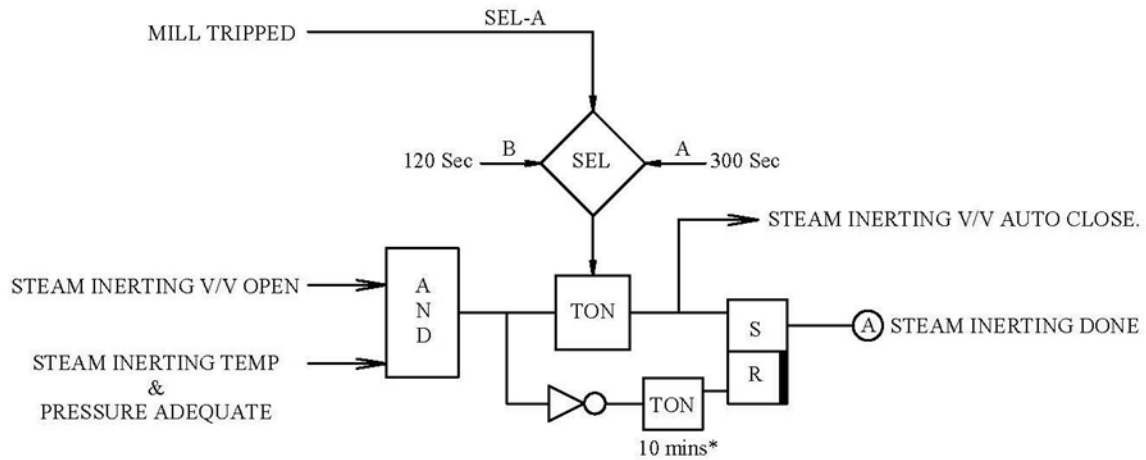
2. ** GAIN VALUE K AND P1 CONTROLLER TO BE TUNED AT SITE CONSIDERING ACTUAL RESPONSE OF PROCESS AND FIELD DEVICES.



NOTES:-

1. * RATE TO BE IMPLEMENTED FOR BUMPLESS TF
2. ** GAIN VALUE K AND P1 CONTROLLER TO BE TUNED AT SITE CONSIDERING ACTUAL RESPONSE OF PROCESS AND FIELD DEVICES.

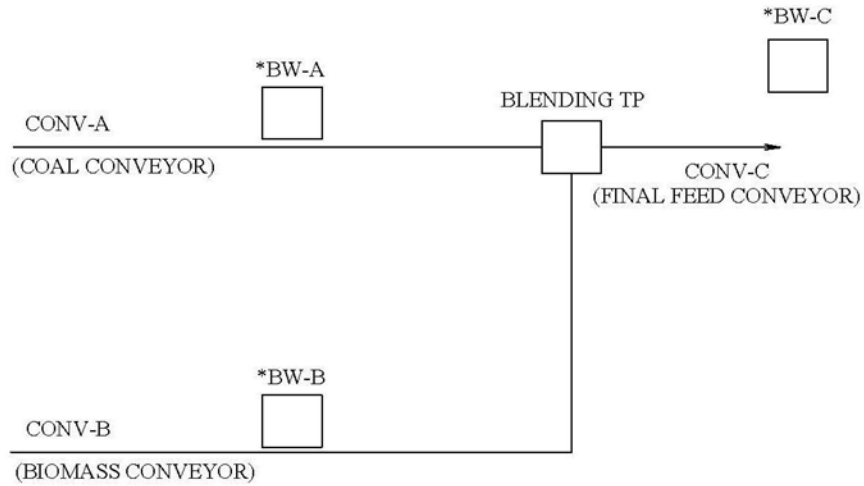
STEAM INERTING LOGIC



NOTES:-

1. * TIMER VALUE ARE SITE TUNABLE.
2. STEAM INERTING VALVE AUTO OPEN CONDITION:
 - (a) MILL TRIPPED

TYPICAL BIOMASS BLENDING SCHEME



1. THIS IS TYPICAL SCHEME INDICATED
2. * BW INDICATED ARE SINGLE / REDUNDANT BASED ON AVAILABILITY
3. SUITABILITY OF BW FOR BIOMASS APPLICATION MAY ENSURED.
4. THE DELAYS ARE CONFIGURABLE BASED ON COMMISSIONING
5. IF AUTOMATIC CONTROLLABLE FEED EQUIPMENT AVIALBLE FOR BIOMASS.THE BIOMASS FEED RATE MAY ALSO BE CONTROLLED.

